GDAŃSK UNIVERSITY OF TECHNOLOGY

Research Methods

Łukasz Sienkiewicz

20-02-2026

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AIMS

The module aims at providing participants with insights and understanding of research methods used in business and management for the purpose of making evidence- based decisions using critical approach and complex information...

...to support drafting the research part of your Capstone projects!

AGENDA

1. Research in managerial decision-making: why do we need evidence-based approach?

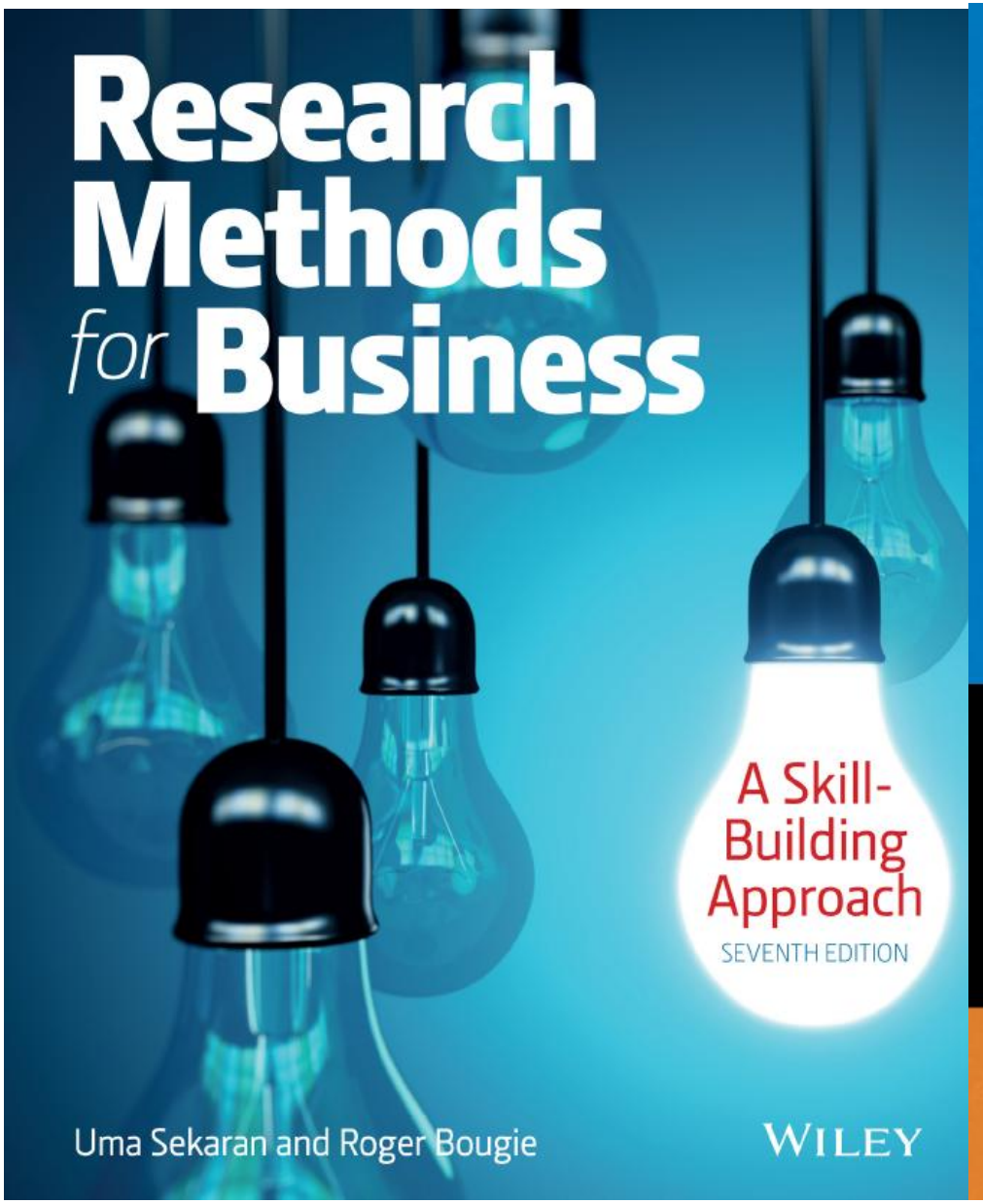
2. Stages of the research process: how to make a research process work in practice?

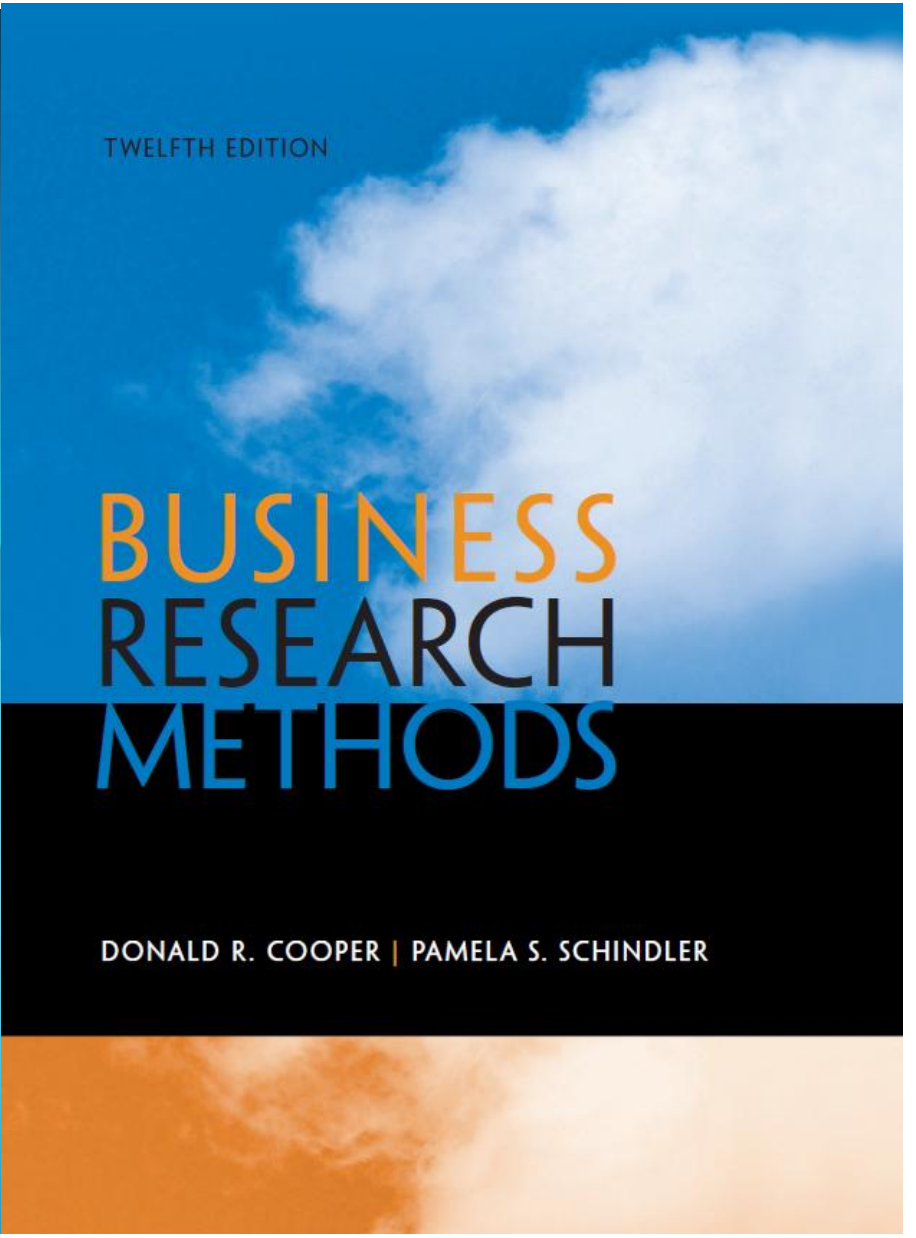
3. Types of business research methods: how to choose fit-for-purpose approach?

RECOMMENDED READINGS

- Cooper D.R., Schindler P.S., *Business Research Methods*, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.
- Bell E., Bryman A., Harley B., *Business Research Methods*, 5th Edition, Oxford University Press, Oxford 2019.
- Sekaran, U., Bougie, R., *Research methods for business : a skill-building approach*, Seventh edition, United Kingdom : John Wiley & Sons, 2016.
- Barends E., Rousseau D.M., *Evidence-Based Management: How to Use Evidence to Make Better Organizational Decisions*, 1st Edition, Kogan Page, 2018.

…or any other textbook on research methods in/for business…


Research Methods for Business
A Skill-Building Approach
SEVENTH EDITION
Uma Sekaran and Roger Bougie
WILEY


TWELFTH EDITION
BUSINESS RESEARCH METHODS
DONALD R. COOPER | PAMELA S. SCHINDLER

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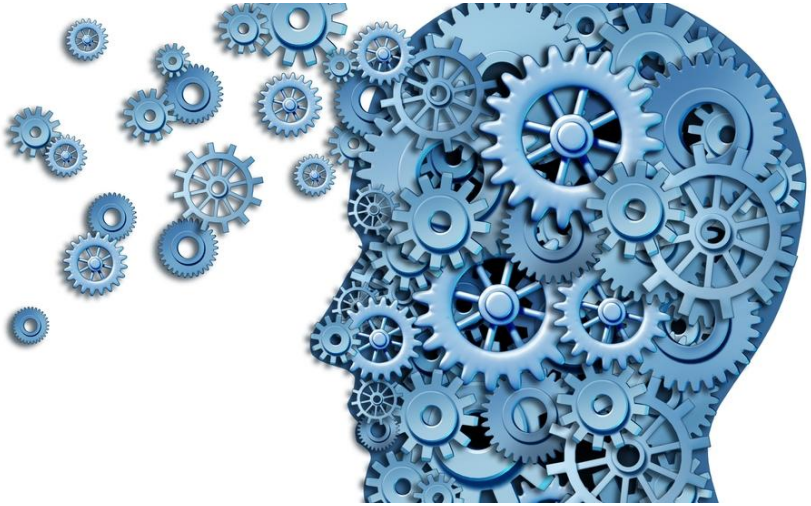
RESEARCH IN MANAGERIAL DECISION-MAKING: WHY DO WE NEED EVIDENCE-BASED APPROACH?

- Rationale for research in managerial work.
- Research methods principles.
- Reasoning models.

Hypothesis:

Leadership and managerial work involves in the first place **rational and accurate (evidence-based) decision making.**



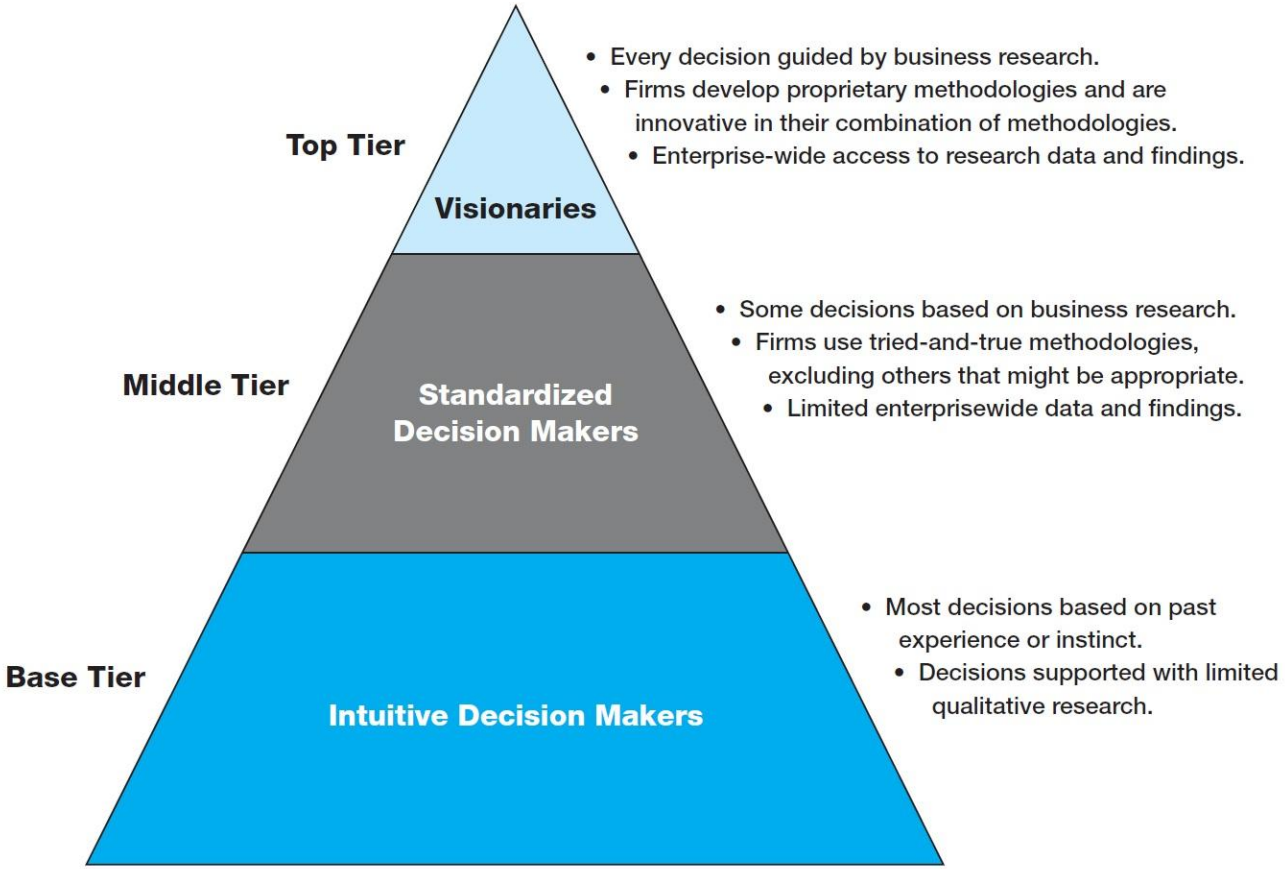
Managerial decisions:

- Intuitive?
- Informed?
- Evidence-based?

Is evidence / research important in leadership role/managerial work?

- Can you think of a situation where you used research in order to inform thinking, decisions, and/or actions in relation to a managerial issue?
- What was the purpose of your research, the problem, the information you collected?
- How did you use this information to support decisions?

Hierarchy of information-based decision makers



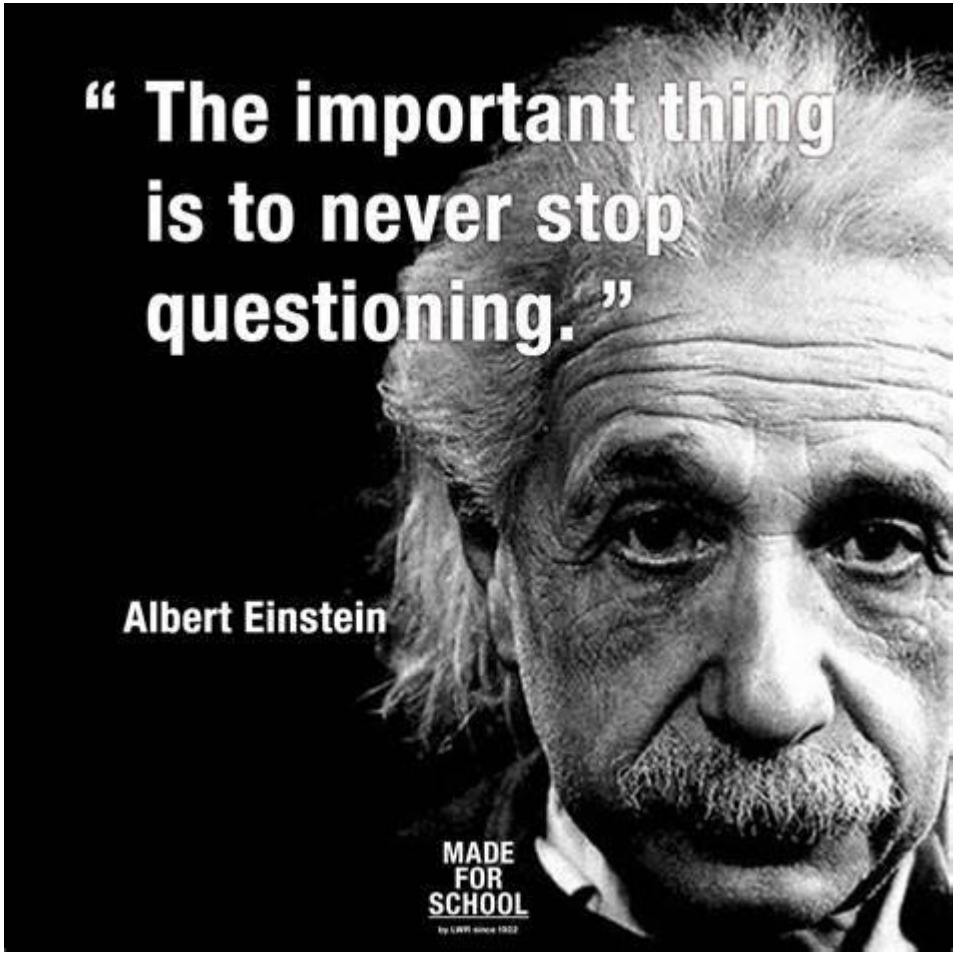
Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

„Most people use statistics in the same way that a drunk uses a lamp-post - for support rather than illumination”

Attributed to Andrew Lang - Scottish poet, novelist, and literary critic (speech of 1910)

THINKING LIKE A RESEARCHER

- If the tools of thinking are the mind of science, then the **scientific attitude** is the spirit.
- The portraits of outstanding scientists are the stories of imagination, intuition, curiosity, suspicion, anguish, the rage to know, and self-doubt.
- A **good business researcher** must also possess these essential predispositions.

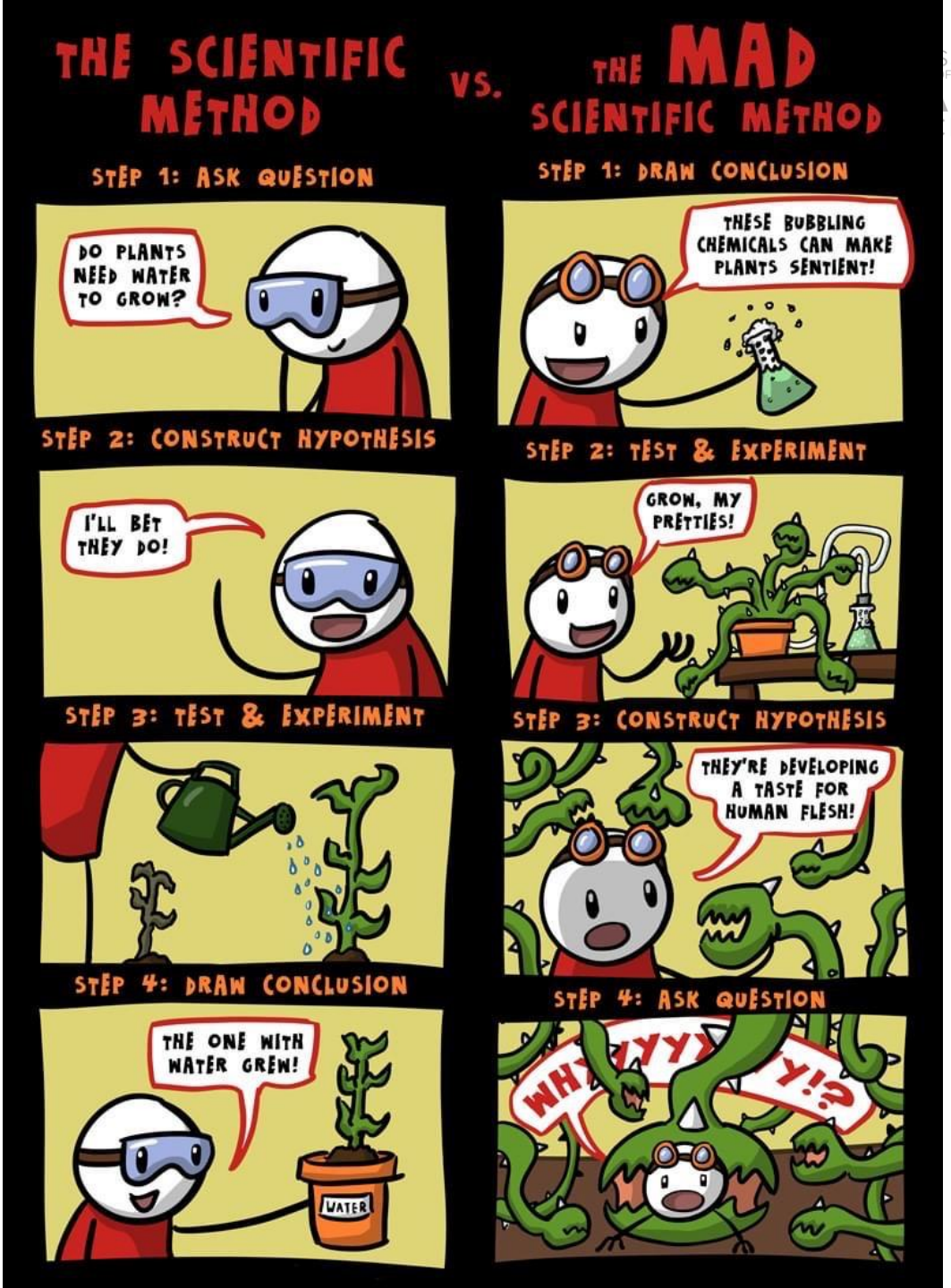


Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

You know nothing, Jon Snow.

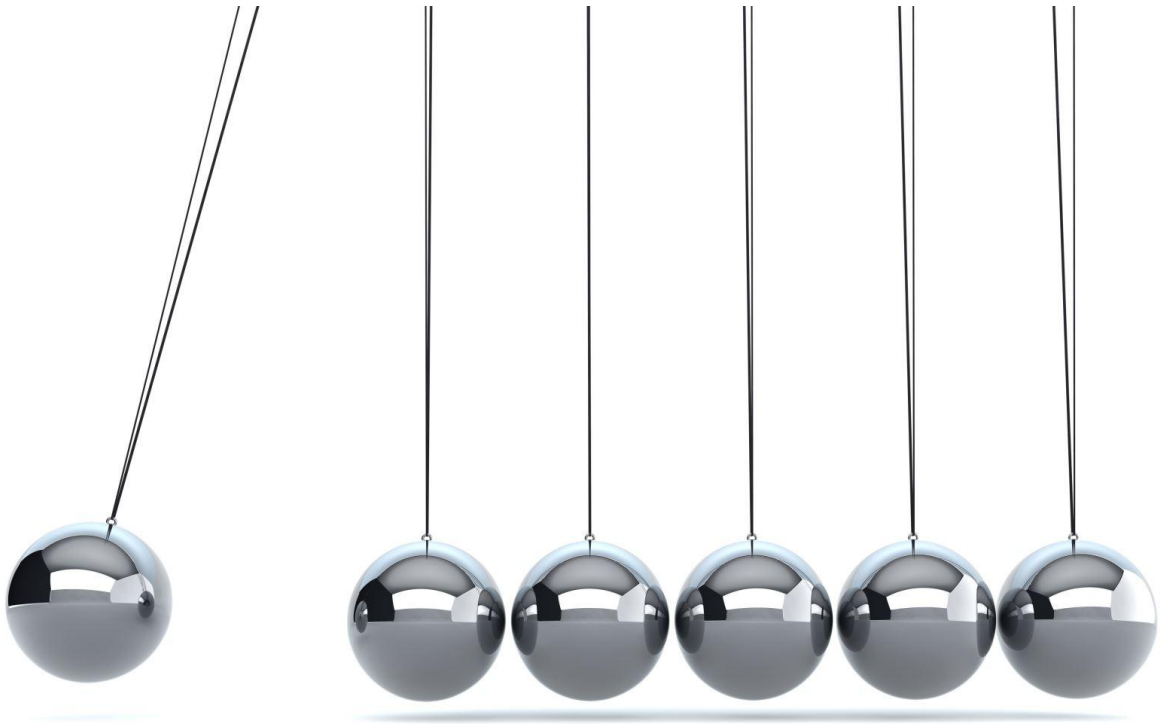
KEY RESEARCH PRINCIPLES

- Scientific method
- Understanding and ability to use data sources
- Research triangulation
- Understanding causal and non-causal relationships
- Research quality and ethics



Good research follows the standards of the **scientific method:**

- systematic,
- problem/hypothesis based
- empirically verifyable procedures for
- generating replicable research.

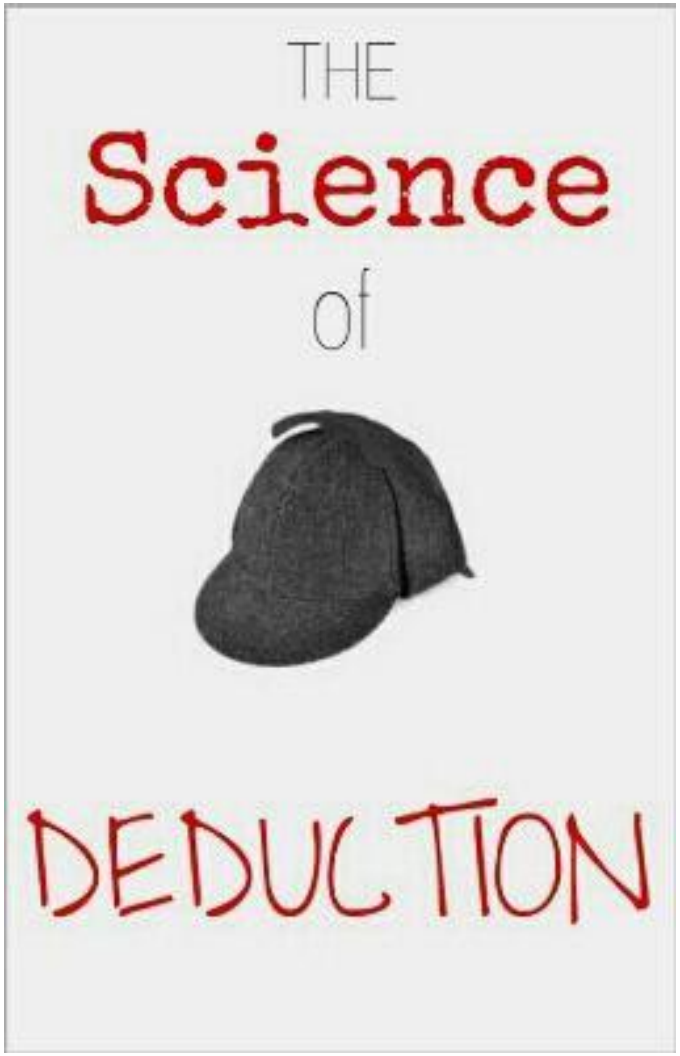


What will happen next?

REASONING MODELS

Deduction (Cambridge Dictionary):

- The process of reaching a decision or answer by thinking about known facts or
- The process of learning something by considering a general set of facts about how something relates to them.
  - *Sherlock Holmes was famous for making clever deductions*



REASONING MODELS

Deduction is a form of argument that purports to be conclusive - the conclusion must necessarily follow from the reasons given.

For a deduction to be correct, it must be both true and valid:

- Premises (reasons) given for the conclusion must agree with the real world (true).
- The conclusion must necessarily follow from the premises (valid).

Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

FINDING PATTERNS

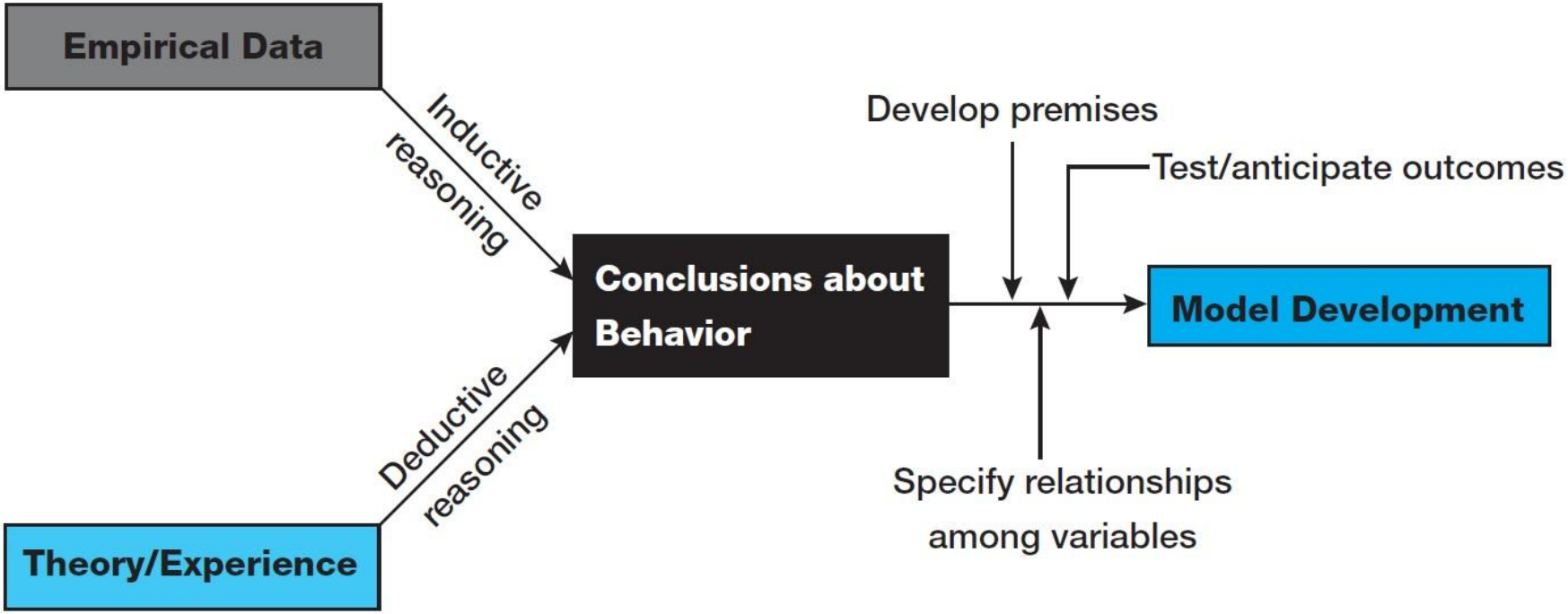
Induction

„In **induction** you draw a conclusion from one or more particular facts or pieces of evidence. The conclusion explains the facts, and the facts support the conclusion.

The nature of induction, however, is that the conclusion is only a hypothesis”.

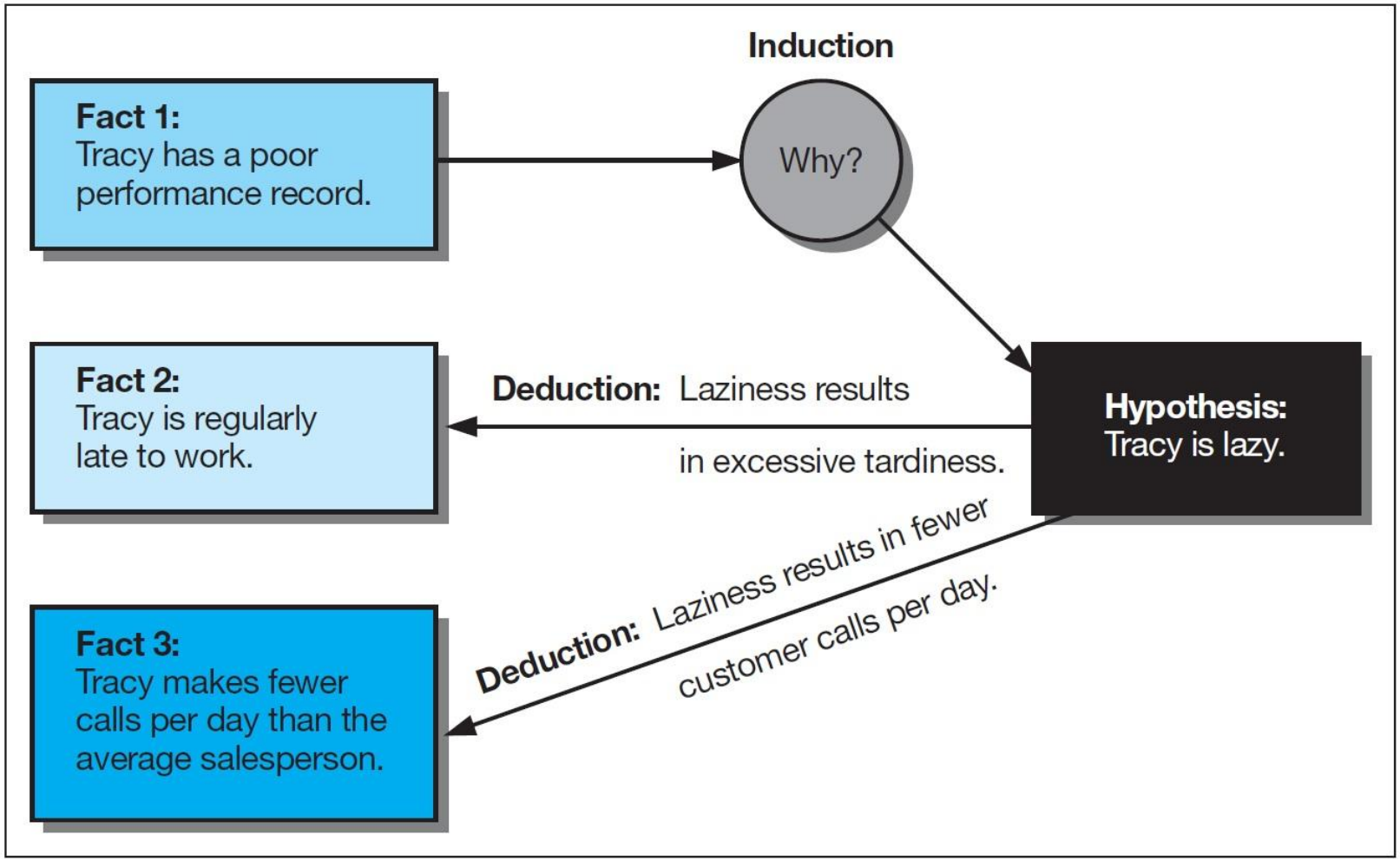
Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

REASONING MODELS



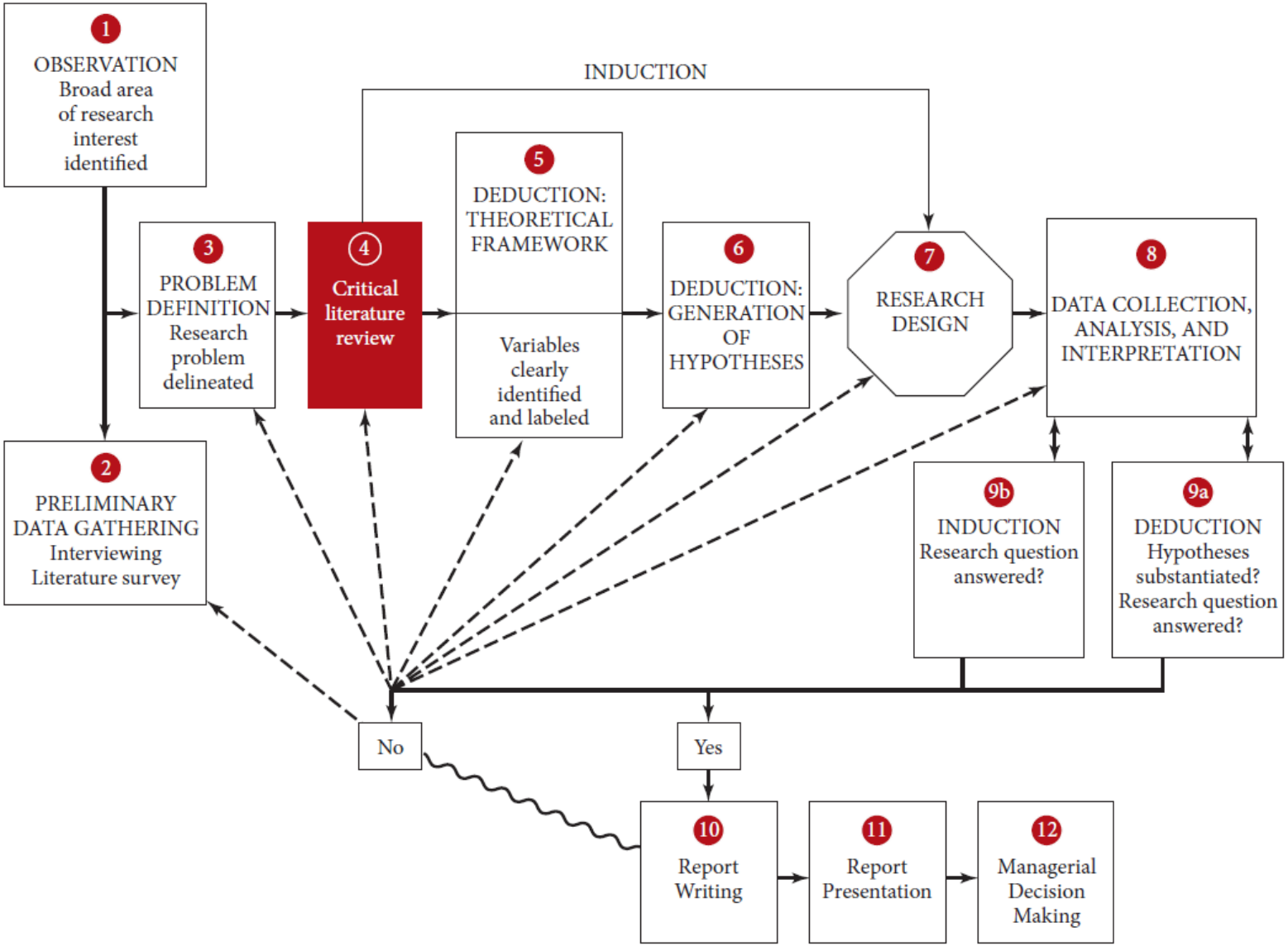
Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

Combining induction and deduction



Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

REASONING MODELS



Source: Sekaran, U., Bougie, R. (2016) *Research methods for business : a skill-building approach*, Seventh edition, United Kingdom : John Wiley & Sons

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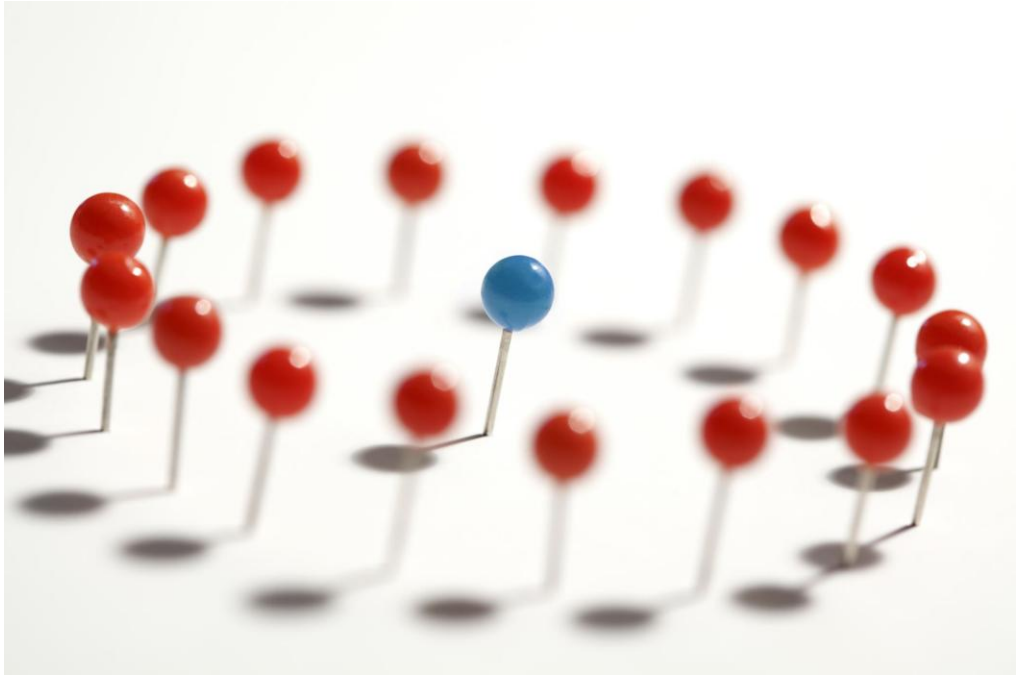
STAGES OF THE RESEARCH PROCESS: HOW TO MAKE A RESEARCH PROCESS WORK IN PRACTICE?



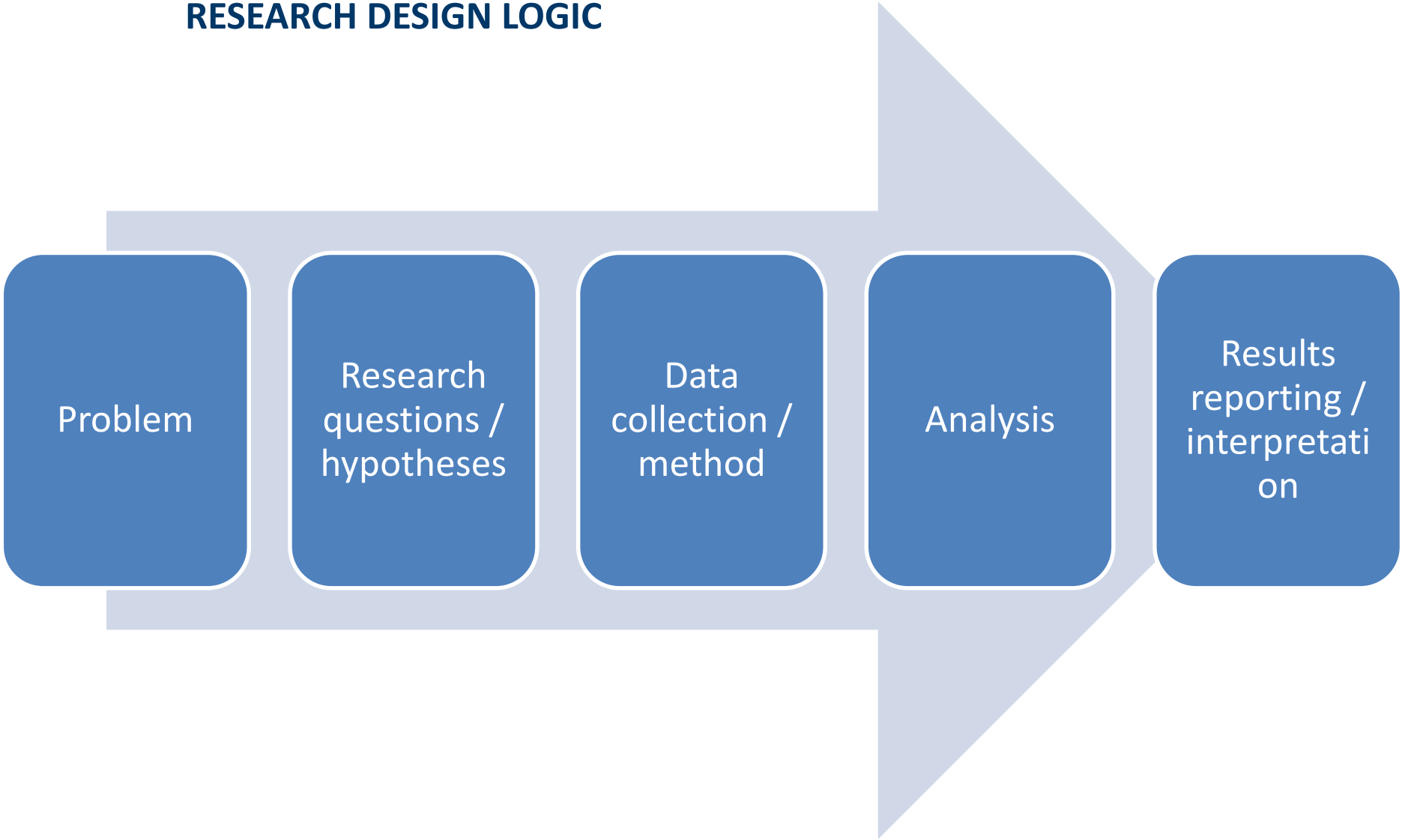
- Problem formulation.
- Research questions.
- Data collection / method.
- Data analysis.
- Results interpretation.

RESEARCH DESIGN

The **research design** is the blueprint for fulfilling objectives and answering questions. Selecting a design may be complicated by the availability of a large variety of methods, techniques, procedures, protocols, and sampling plans.

Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

RESEARCH DESIGN LOGIC



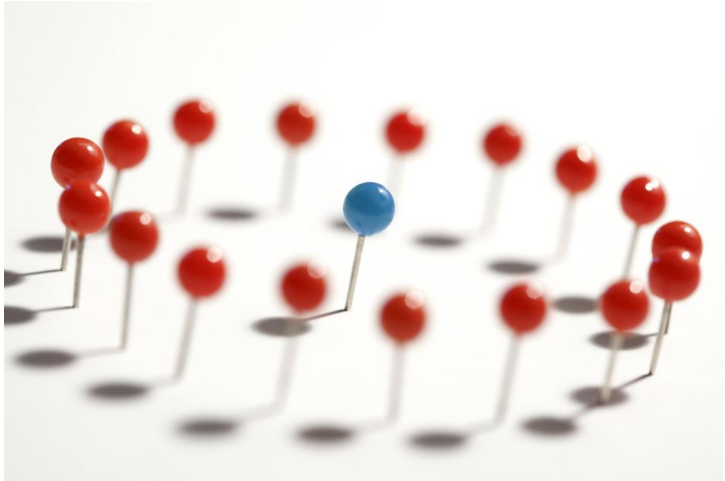
DILEMMA / PROBLEM

Problem (or dilemma) formulation is the process of clearly defining the managerial issue or decision challenge that the research aims to address, ensuring that the study is focused, relevant, and actionable.

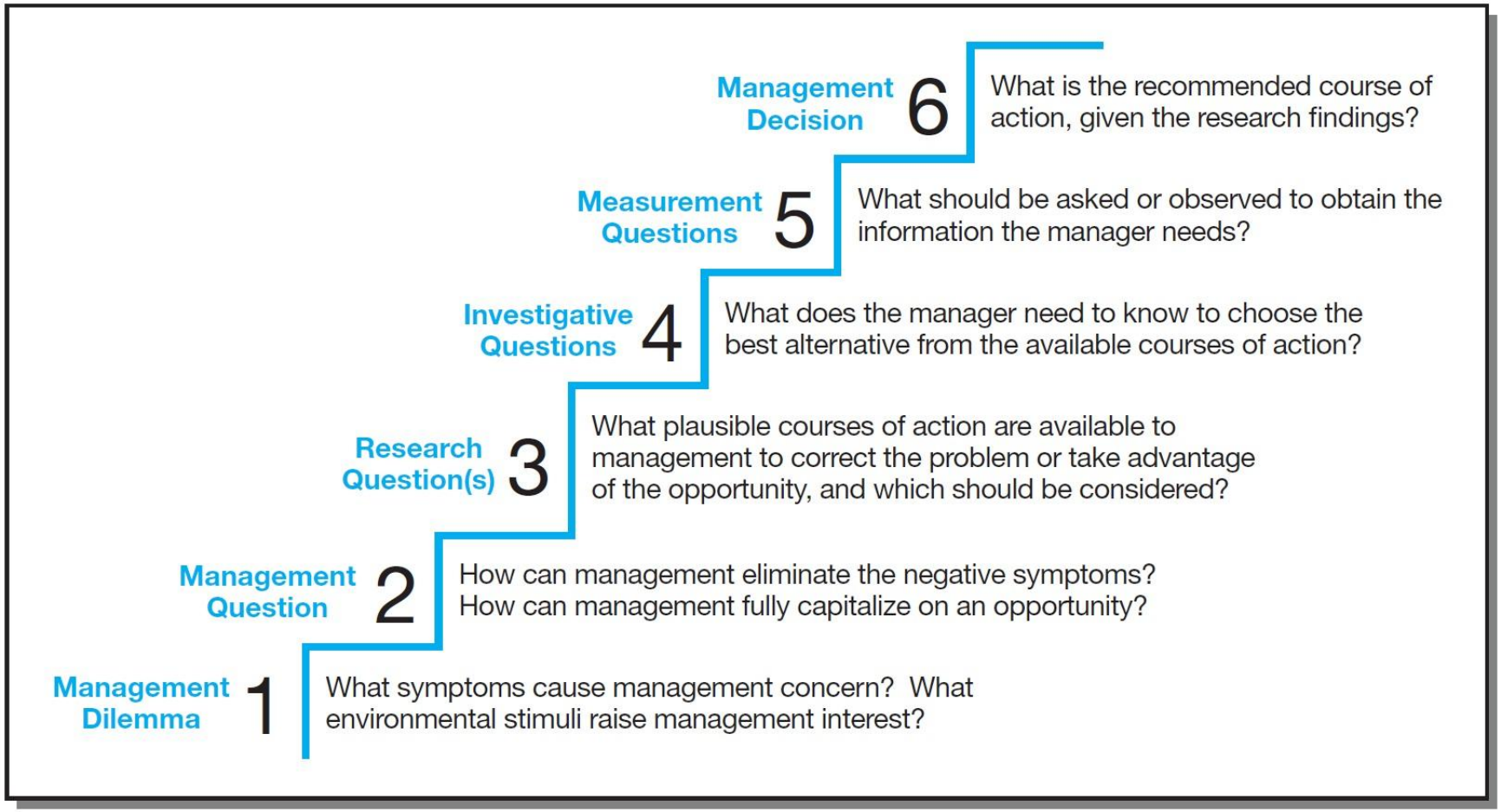
The process begins at the most general level with the **dilemma / problem.**

For business research this is usually a symptom of an actual problem, such as:

- Rising costs.
- Declining sales.
- Development of a new technology / product / service.
- Competitors activity.
- Personnel / HRM issues.
- …



Management-research question hierarchy

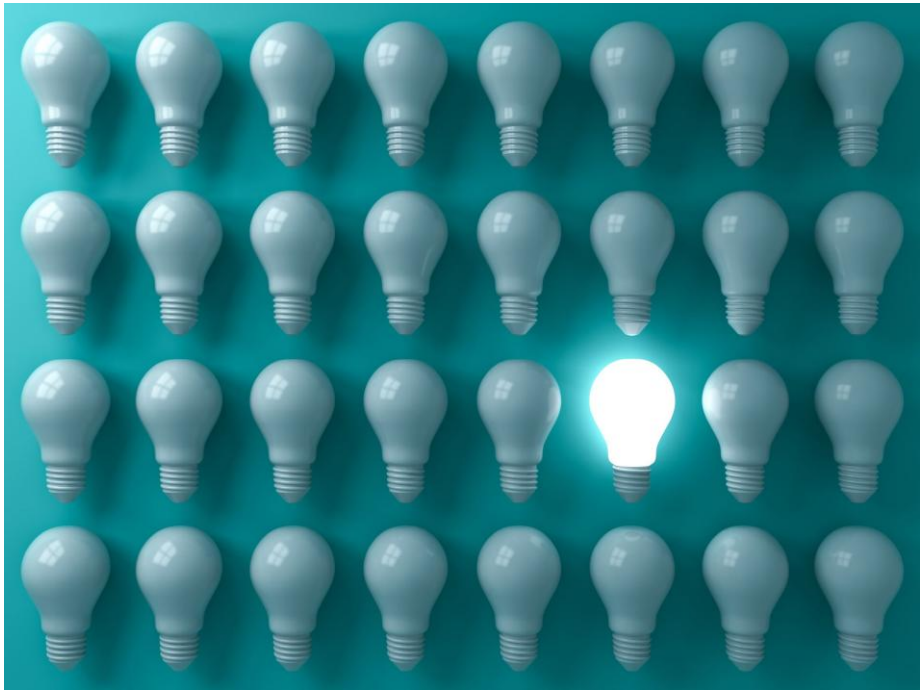


Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

HYPOTHESIS

We define a **proposition** as a statement about observable phenomena (concepts) that may be judged as true or false.

When a proposition is formulated for empirical testing, we call it a **hypothesis.**



Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

RESEARCH QUESTIONS / HYPOTHESES

Descriptive hypotheses state the existence, size, form, or distribution of some variable. Business researchers often use a **research question** rather than a descriptive hypothesis.

Descriptive Hypothesis Format	*Research Question Format*
In Detroit (case), our potato chip market share (variable) stands at 13.7 percent.	What is the market share for our potato chips in Detroit?
American cities (cases) are experiencing budget difficulties (variable).	Are American cities experiencing budget difficulties?
Eighty percent of Company Z stockholders (cases) favor increasing the company's cash dividend (variable).	Do stockholders of Company Z favor an increased cash dividend?
Seventy percent of the high school–educated males (cases) scavenge in the Army firing range for salvageable metals (variable).	Do a majority of high school–educated male residents scavenge in the Army firing range for salvageable metals?

Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

RESEARCH QUESTIONS / HYPOTHESES

The research question format is less frequently used with a situation calling for **relational hypotheses.**

Correlational hypotheses state that the variables occur together in some specifi ed manner without implying that one causes the other.

Example: „Young women (under 35 years of age) purchase fewer units of our product than women who are 35 years of age or older".

With **explanatory (causal) hypotheses,** there is an implication that the existence of or a change in one variable causes or leads to a change in the other variable. As we noted previously, the causal variable is typically called the independent variable (IV) and the other the dependent variable (DV).

Example: „An increase in family income (IV) leads to an increase in the percentage of income saved (DV)."

Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

HOW TO FORMULATE GOOD RESEARCH QUESTIONS FOR AN MBA PROJECT

A good research question translates a managerial problem into a clear, focused, and researchable inquiry that guides the choice of methods and data.

Characteristics of good research questions:

- clearly linked to a real managerial problem or decision
- specific and focused (not too broad or vague)
- empirically researchable (data can be realistically collected)
- neutral and unbiased (not leading to a predetermined answer)
- relevant for managerial recommendations and decision-making

Good practice tips:

- start from the business problem → then derive the research question
- use precise concepts and clearly defined variables
- check feasibility: access to data, respondents, and time constraints
- ensure alignment with chosen research method (survey, interviews, mixed methods)

DATA COLLECTION DESIGN

Data collection design defines what data are needed, from which sources, and by which methods to answer the research questions in a valid, reliable, and feasible way.

Types of information sources:

- **Primary sources** – original data collected directly for the study (e.g., surveys, interviews, company data)
- **Secondary sources** – interpretations or analyses of primary data (e.g., academic articles, reports, textbooks, media)
- **Tertiary sources** – tools that help locate and organize information (e.g., indexes, bibliographies, search engines)

Common data sources & collection methods in MBA Projects:

- Desk research: analysis of existing reports, statistics, and organizational data
- Surveys and questionnaires (quantitative methods)
- Interviews and focus groups (qualitative methods)
- Case study materials and document analysis

WHAT IS LITERATURE REVIEW?

A literature review is a structured synthesis and critical evaluation of existing academic and empirical studies relevant to a research problem, used to identify what is already known, what gaps remain, and how the current study contributes to evidence-based understanding.

Purpose of review in MBA projects:

- to understand what is known / avoid reinventing the wheel
- to identify gaps
- to build conceptual framework

TYPES OF LITERATURE REVIEW

Type	Purpose	Typical use
Narrative review	Broad overview of topic	Exploratory MBA thesis
Systematic review	Rigorous structured search	Evidence-based policy/business
Scoping review	Mapping existing research	New or emerging topic
Critical review	Evaluate strengths & weaknesses	Theory-focused thesis
Meta-analysis	Statistical synthesis of studies	Quantitative evidence integration

WHICH TYPES OF LITERATURE REVIEW ARE MOST COMMON IN MBA / CAPSTONE PROJECTS

Narrative literature review

- **Why used in MBA projects:**
  - Provides a structured overview of key theories, models, and prior studies
  - Flexible and feasible within limited time
  - Supports development of conceptual framework for applied research
- **Typical use:**
  Explaining what is already known about a managerial problem (e.g., employee engagement, digital transformation, leadership styles).

Critical literature review

- **Why used in MBA projects:**
  - Goes beyond summarizing → evaluates strengths and weaknesses of existing research
  - Helps justify chosen approach and research gap
  - Useful when comparing competing managerial concepts or models
- **Typical use:**
  Assessing different frameworks (e.g., performance management models) and selecting the most suitable one for the organization.

WHAT IS DESK RESEARCH?

Research based on existing data, reports, and empirical studies (not primary data collection).

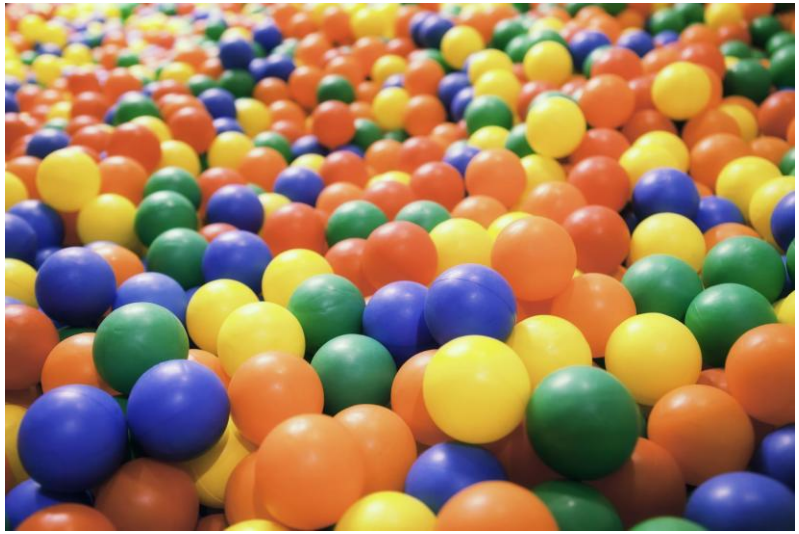
Literature review	Desk research
Focus on theory and prior studies	Focus on empirical evidence & data
Academic emphasis	Managerial decision emphasis
Conceptual framework	Evidence-based diagnosis

Examples of sources:

- Academic empirical studies (Scopus, Web of Science, Google Scholar)
- Industry reports (e.g. McKinsey, Delloitte, sectoral reports)
- Governmental and market statistics (Eurostat, GUS, statista)
- Company internal data

METHOD & SAMPLING DESIGN

Another step is to identify the **target population** (those people, events, or records that contain the desired information and can answer the measurement questions) and then determine whether a **sample** or a **census** is desired.



Taking a **census** requires that the researcher examine or count all elements in the target population. A **sample** examines a portion of the target population, and the portion must be carefully selected to represent that population.

Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

INTRODUCTION TO SAMPLING

Sampling is the process of selecting a subset of individuals, cases, or observations from a larger population in order to draw conclusions about that population in a feasible and systematic way.

Probability sampling	Non-probability sampling
Random selection	Non-random selection
Generalizable results	Exploratory insights
Quantitative research	Qualitative / case studies

Why sampling matters: You rarely study the whole population → sampling determines validity of conclusions.

PROBABILITY SAMPLING

Probability sampling is a sampling approach in which every unit in the population has a known and non-zero chance of being selected, enabling statistical generalization of results.

Key characteristics:

- random selection of units
- known probability of inclusion
- enables statistical inference and generalization
- typically used in quantitative research

When appropriate in MBA Projects:

- survey research aiming to generalize findings
- large and well-defined populations (e.g., employees, customers)
- when representativeness is crucial for decision-making

PROBABILITY SAMPLING

Sampling method	Short description	Typical MBA use example
Simple random sampling	Every unit has an equal chance of selection	Survey of randomly selected employees across the organization
Stratified sampling	Population divided into subgroups and sampled proportionally	Survey comparing managers vs. non-managers or different departments
Cluster sampling	Random selection of naturally occurring groups (clusters)	Selecting several branches/units and surveying all employees within them
Systematic sampling	Selecting every nth unit from an ordered list	Selecting every 10th customer from a CRM database

NON-PROBABILITY SAMPLING

Non-probability sampling is a sampling approach in which units are selected based on accessibility, judgment, or specific criteria rather than random selection, limiting statistical generalization.

Key characteristics:

- non-random selection of participants
- probability of inclusion is unknown
- suitable for exploratory and qualitative research
- focus on depth and relevance rather than representativeness

When appropriate in MBA Projects:

- case studies and qualitative interviews
- exploratory research on new or complex problems
- limited access to full population lists
- research focusing on experts or specific stakeholder groups

NON-PROBABILITY SAMPLING

Sampling method	Short description	Typical MBA use example
Convenience sampling	Selection of easily accessible participants	Survey among available employees or students
Purposive sampling	Selection of participants based on relevance or expertise	Interviews with key managers or industry experts
Snowball sampling	Participants recruit additional relevant participants	Research on niche professional groups or networks
Quota sampling	Selection to match key characteristics of the population	Ensuring equal number of respondents from different departments or roles

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TYPES OF BUSINESS RESEARCH METHODS: HOW TO CHOOSE FIT-FOR-PURPOSE APPROACH?

- Qualitative methods.
- Quantitative methods.
- Mixed & other research methods.

QUALITATIVE METHODS

Qualitative research – a type of research that aims to find out people's opinions and feelings rather than information that can easily be shown in numbers (Cambridge Dictionary)

Qualitative research includes an "array of interpretive techniques which seek to describe, decode, translate, and otherwise come to terms with the meaning, not the frequency, of certain more or less naturally occurring phenomena in the social world." (Cooper & Schindler)

IN-DEPTH INTERVIEWS (IDI)

A type of qualitative research involving an unstructured personal interview with a single respondent, conducted by a highly skilled interviewer.

The purpose of in-depth interviews is to understand the underlying motivations, beliefs, attitudes, and feelings of respondents on a particular subject.

Source: [In-Depth Interview | Insights Association]

IN-DEPTH INTERVIEWS (IDI)

A type of qualitative research involving an unstructured personal interview with a single respondent, conducted by a skilled interviewer. The purpose of in-depth interviews is to understand the underlying motivations, beliefs, attitudes, and feelings of respondents on a particular subject.

IDI types

- **unstructured interview** - no specific questions or order of topics to be discussed, with each interview customized to each participant; generally starts with a participant narrative,
- **semistructured interview** - generally starts with a few specific questions and then follows the individual's tangents of thought with interviewer probes,
- **structured interview** - often uses a detailed interview guide similar to a questionnaire to guide the question order and the specific way the questions are asked, but the questions generally remain open-ended.

IN-DEPTH INTERVIEWS (IDI)

When to use IDIs in MBA Projects:

- Exploring complex organizational issues (e.g., leadership, culture, change management)
- Understanding perspectives of key stakeholders or experts
- Investigating sensitive topics where privacy encourages openness
- Complementing survey results with deeper explanations.

Strengths:

- rich, detailed and context-specific insights
- flexibility to probe and clarify responses
- useful for understanding underlying causes and mechanisms

Limitations:

- time-consuming data collection and analysis
- limited generalizability due to small samples
- potential interviewer bias

FOCUS GROUP INTERVIEWS (FGI)

A type of qualitative research that consists of an informal discussion of a particular topic with a small number of selected participants (usually 6-10). The discussion is guided by a moderator who does not influence the outcome, but ensures that all the subject areas are discussed by the group and the views of the participants are as clear as possible.

FGI types:

- **Homogeneous groups** – participants with similar roles or backgrounds (e.g., only managers or only frontline staff)
- **Heterogeneous groups** – diverse participants to capture multiple perspectives
- **Mini focus groups** – smaller groups (3–4 participants) for more in-depth discussion
- **Online focus groups** – virtual discussions using digital platforms (useful for dispersed teams)

FOCUS GROUP INTERVIEWS (FGI)

When to use FGIs in MBA Projects:

- Exploring perceptions of employees or customers about policies or initiatives
- Generating ideas, reactions, and collective feedback
- Understanding group norms and organizational climate
- Early-stage exploratory research before designing a survey

Strengths:

- Interaction generates diverse perspectives and new insights
- Efficient way to collect multiple viewpoints simultaneously
- Useful for identifying common themes and disagreements

Limitations:

- Risk of dominant participants influencing discussion
- Less suitable for sensitive or confidential topics
- Requires strong moderation to keep discussion focused

CASE STUDIES

A case study is an empirical method that investigates a contemporary phenomenon in its real-life context, especially when the boundaries between the phenomenon and its context are not clearly evident and multiple sources of evidence are used.

Type	Main Purpose	Typical Focus
Exploratory	Investigate a new or insufficiently understood phenomenon	Identifying key issues, patterns, and future research questions
Explanatory	Test hypotheses and explain causal mechanisms	Understanding “how” and “why” outcomes occur
Descriptive / Example	Describe features of good practice or specific processes	Providing clear, practice-oriented case descriptions

CASE STUDIES

When to use FGIs in MBA Projects:

- For studying complex managerial or organizational problems
- When context (culture, structure, environment) strongly influences outcomes
- When boundaries between phenomenon and context are not clearly defined

Strengths:

- Rich, in-depth and context-sensitive insights
- Strong practical relevance for real managerial decisions
- Allows integration of qualitative and quantitative evidence (triangulation)
- Useful for exploring new issues or explaining organizational processes

Limitations:

- Limited statistical generalizability beyond the case
- Risk of researcher subjectivity and interpretation bias
- Time-consuming data collection and analysis
- Requires clear justification of case selection and evidence sources

Quantitative research focuses on the systematic measurement and numerical analysis of variables to identify patterns, relationships, and differences relevant to managerial decision-making.

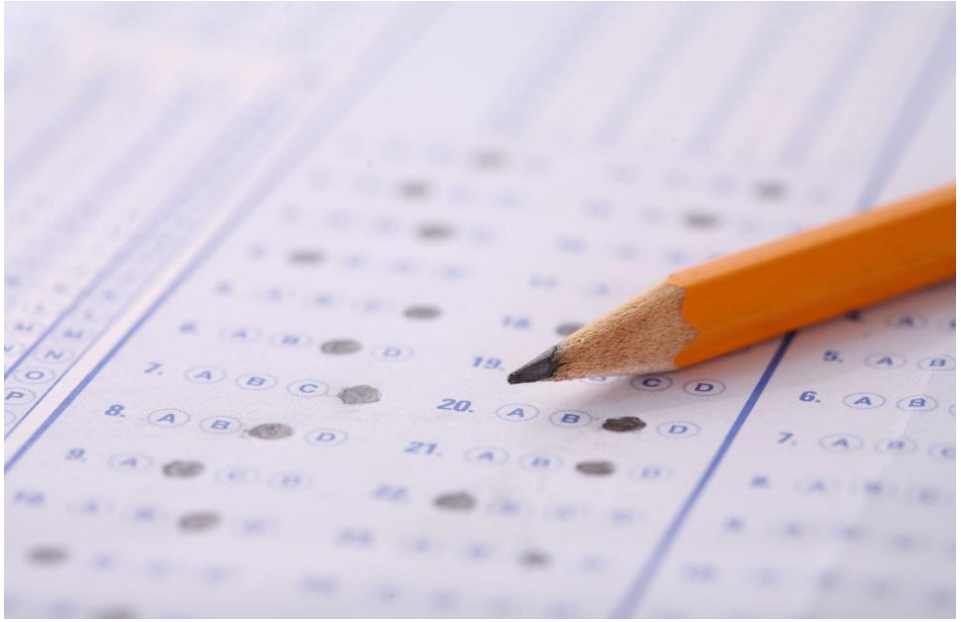
In MBA Projects, quantitative research typically measures:

- Employee attitudes, engagement, and satisfaction
- Customer behaviour, preferences, and loyalty
- Performance indicators and organizational outcomes
- Relationships between managerial practices and results

Key characteristics:

- Structured and standardized data collection
- Use of numerical data and statistical analysis
- Focus on testing hypotheses and identifying relationships
- Possibility of comparing groups or tracking trends over time

SURVEYS



A survey is a structured data collection method using standardized questions to obtain comparable information from a defined group of respondents.

Key features:

- Same set of questions asked to all participants
- Enables comparison across individuals, teams, or departments
- Can be administered online, by phone, or in person
- Suitable for measuring perceptions, behaviors, and attitudes

SURVEYS

Type of Survey	Description	Advantages	Limitations
Self-administered survey (CAWI / online)	Respondents complete a standardized questionnaire independently (e.g., online forms, intranet surveys)	Low cost, fast data collection, large samples possible	Risk of low response rate, limited control over understanding of questions, limited control over sample effective structure
Telephone survey (CATI)	Structured questionnaire conducted via phone by an interviewer	Higher response rates than online, possibility to clarify questions	More time-consuming and costly, potential interviewer influence
Pen&Paper survey (PAPI)	Respondents complete a printed questionnaire distributed in person (Paper-and-Pencil Interviewing)	High control over distribution, suitable in organizational settings, useful where digital access is limited	Manual data entry required, risk of missing data, more administrative effort

MIX METHODS RESEARCH

Mixed methods research integrates quantitative (e.g., surveys, statistical data) and qualitative (e.g., interviews, case studies) approaches within one study to gain a more comprehensive understanding of a managerial problem.

When to use mixed methods:

- When both measurement and deeper understanding are needed
- When survey results require explanation or contextualization
- When exploring a new issue and then validating findings quantitatively
- When addressing complex organizational problems involving both behaviors and perceptions

Advantages:

- triangulation: cross-verifying findings from multiple data sources
- richer insights: combines numerical trends with in-depth explanations
- higher practical relevance for managerial decision-making
- reduces limitations of relying on a single method

SWOT ANALYSIS

- SWOT analysis is a strategic planning technique that provides assessment tools.
- Identifying core strengths, weaknesses, opportunities, and threats lead to fact-based analysis, fresh perspectives and new ideas.
- SWOT analysis works best when diverse groups or voices within an organization are free to provide realistic data points rather than prescribed messaging.

PEST(LE) ANALYSIS

- This type of analysis is used to gauge external factors that could impact the profitability of a company.
- Generally, it is more effective with larger organizations that are more likely to experience the effects of macro events.
- PEST(LE) analysis is commonly used in conjunction with SWOT analysis

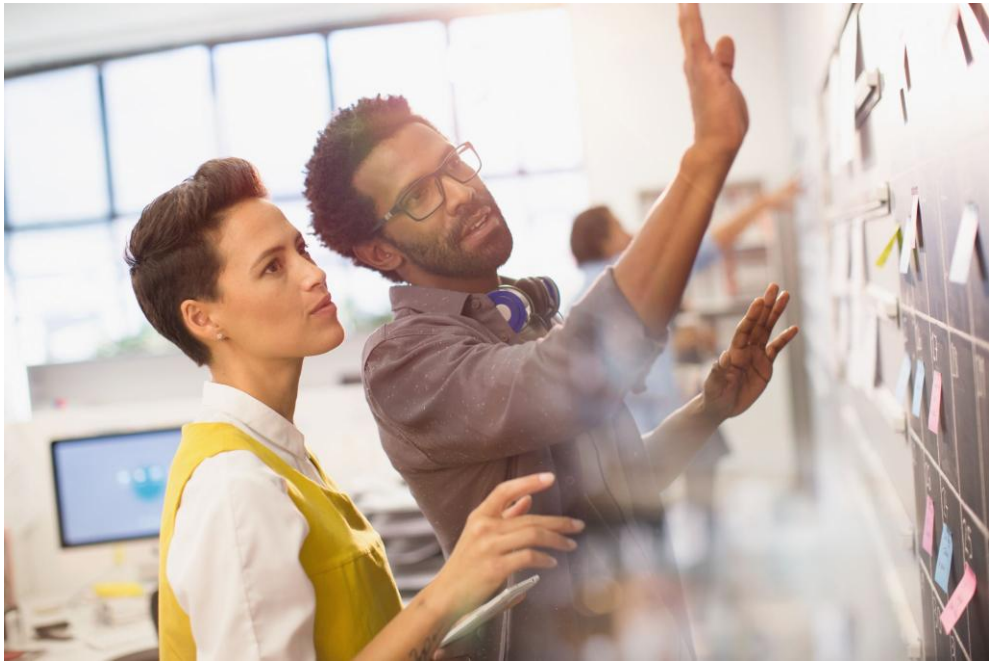
Source: investopedia.com

OTHER METHODS

- Observation studies
- Experience / action research
- Experimental designs

PILOT TESTING

A **pilot test** is conducted to detect weaknesses in design and instrumentation and to provide proxy data for selection of a probability sample. It should, therefore, draw subjects from the target population and simulate the procedures and protocols that have been designated for data collection.



Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

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RESEARCH VALIDITY AND RELIABILITY

Validity and reliability are key criteria of research quality, ensuring that a study measures the right concepts and produces consistent, trustworthy results that can support sound managerial decisions.

Validity: are we measuring the right things?

- Does the research truly capture the intended concept?
- Are survey questions aligned with the research problem?
- Are conclusions logically supported by the data?

- **Example:**
  If the goal is to measure "employee engagement," but the survey only measures job satisfaction, validity is limited.

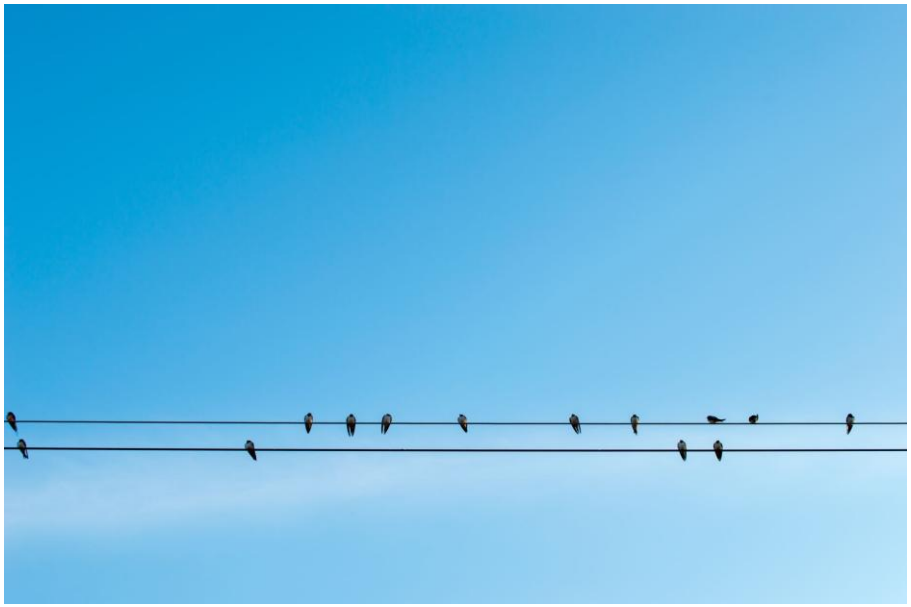
Reliability: are the results consistent and stable?

- Would the study produce similar results if repeated?
- Are measurement tools clear and standardized?
- Are responses stable across time or groups?

- **Example:**
  If different respondents interpret the same question differently, reliability is low.

FIELDWORK

Fieldwork can mean the data collection stage of a project; or how researchers go about collecting data; or more narrowly, data collection in a social setting that tries to reflect the naturally occurring order of events and subjective meanings of those being studied.



Source: Key Concepts in Social Research

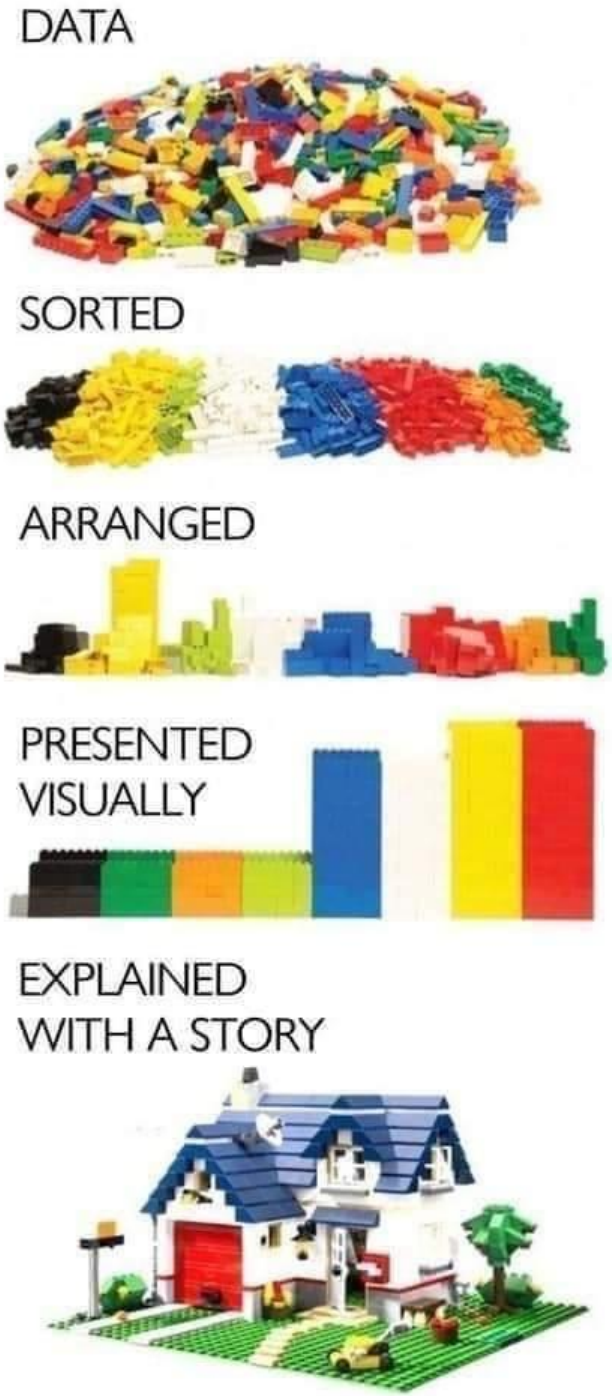
FIELDWORK

Key ethical principles in fieldwork:

- Be transparent about what data are collected, for what purpose, and how they will be used and shared
- Protect all personal data from unauthorized access and disclosure
- Ensure participation is voluntary and based on informed consent
- Avoid any actions that may harm participants or damage their reputation or well-being

Key definitions for MBA research:

- **Consent:** Freely given and informed agreement to participate and provide personal data
- **Data subject:** Any individual whose personal data are used in the research
- **Personal data:** Any information that can identify a person directly or indirectly (e.g., name, email, role, recordings)
- **Harm:** Physical, financial, reputational, or privacy-related negative consequences for participants



DATA ANALYSIS

Data analysis usually involves reducing accumulated data to a manageable size, developing summaries, looking for patterns, and applying statistical techniques. Scaled responses on questionnaires and experimental instruments often require the analyst to derive various functions, as well as to explore relationships among variables.

Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

DATA ANALYSIS

Data analysis involves systematically examining collected data to identify patterns, relationships, and meanings that answer the research questions and support managerial decision-making.

Key Principles:

- Analysis must follow research questions and objectives
- Analytical methods should match the type of data (quantitative or qualitative)
- Transparency and logical reasoning are essential
- Results should lead to actionable managerial conclusions

In MBA Projects:
Data analysis should not only describe results but explain their implications for organizational decisions and recommendations.

QUANTITATIVE DATA ANALYSIS

Main goal:
To analyze numerical data in order to measure relationships, differences, and trends relevant to managerial problems.

Key methods in MBA Projects:

- Descriptive statistics (means, percentages, distributions)
- Cross-tabulations (comparisons between groups)
- Correlation analysis (relationships between variables)
- Regression analysis (identifying key drivers and predictors)
- Trend and comparative analysis over time or units

QUALITATIVE DATA ANALYSIS

Main Goal:
To interpret non-numerical data (e.g., interview transcripts, documents) in order to understand meanings, perceptions, and organizational processes.

Key methods in MBA Projects:

- Thematic analysis (identifying recurring themes and patterns)
- Content analysis (systematic categorization of textual data)
- Coding and categorization of responses
- Narrative analysis (understanding stories and experiences)
- Comparative case analysis (identifying similarities and differences across cases)

INTEGRATING QUANTITATIVE AND QUALITATIVE ANALYSIS (MIXED APPROACH)

Why combine both types of analysis?

- Quantitative data show “what” and “how much”
- Qualitative data explain “why” and “how”
- Together they provide a more complete understanding of managerial problems

Principle of triangulation:

Using multiple data sources and methods to validate findings and strengthen conclusions.

Typical MBA application:

- Survey identifies low engagement levels (quantitative)
- Interviews explain underlying causes (qualitative)
- Combined analysis supports targeted managerial recommendations

REPORTING & USING RESULTS

Finally, it is necessary to transmit the findings and recommendations to the manager for the intended purpose of decision making.

The researcher adjusts the style and organization of the report according to the target audience, the occasion, and the purpose of the research.

Source: Cooper D.R., Schindler P.S., Business Research Methods, Twelfth Edition, McGraw-Hill/Irwin, New York 2014.

Key principles of interpretation:

- Link results directly to research questions and objectives
- Explain not only *what* was found, but also *why it matters* for the organization
- Integrate quantitative results (patterns, relationships) with qualitative insights (causes, explanations)
- Consider context, limitations, and alternative interpretations of findings

Good practices in reporting results:

- Present key findings clearly using tables, charts, and concise summaries
- Highlight the most important insights rather than all results
- Distinguish between factual findings and managerial interpretation
- Formulate evidence-based recommendations aligned with the research evidence

AI IN RESEARCH

AI can significantly support the research process, but it does not replace methodological reasoning, critical evaluation, or academic integrity.

AI can support:

- Refining research problems and questions
- Searching and screening literature
- Summarizing articles and reports
- Generating interview questions or survey drafts
- Supporting qualitative coding and text analysis
- Improving clarity of academic writing

AI cannot:

- Define the research problem for you
- Ensure validity and reliability
- Replace critical interpretation of results
- Guarantee factual accuracy of sources

AI ACROSS THE RESEARCH PROCESS

Research stage	How AI can help	Researcher's responsibility
Problem formulation	Clarify topic, suggest research questions	Ensure relevance and feasibility
Literature review	Identify themes, summarize articles	Verify sources and interpretations
Research design	Suggest methods and instruments	Ensure methodological fit
Data analysis	Support coding, pattern detection	Interpret results critically
Writing & reporting	Improve clarity and structure	Maintain originality and accuracy

AI IN LITERATURE REVIEW

Potential uses:

- Generating keyword lists and search strategies
- Summarizing academic articles and reports
- Identifying key themes and theoretical perspectives
- Comparing findings across studies
- Mapping research gaps

Important risks:

- Hallucinated or non-existent references
- Oversimplified summaries
- Bias due to training data limitations
- Missing recent or niche sources

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Elicit

VOSviewer
Visualizing scientific landscapes

COMMON MISTAKES IN MBA RESEARCH (AND HOW TO AVOID THEM)

Frequent pitfalls	Good practice
• Collecting data without a clear research question • Using samples that are too small or biased • Over-reliance on a single method (only survey or only interviews) • Confusing literature review with personal opinions or unverified sources • Using AI-generated content or references without verification	• Start with a clear managerial problem and research question • Choose methods that fit the problem (not convenience) • Combine data sources where possible (triangulation) • Critically review credible academic and empirical evidence • Verify all AI-supported outputs and references before use

NEXT STEPS…

- Each Capstone project team should prepare **Research Proposal Outline** for their project – please use the structure as presented in the separate file
- Each team presents the outline during online consultations (around 30 minutes for each project)
- All of other teams are welcome to actively participate, with following discussions / possible changes
- Dates for consultations proposed by MBA office
- This is **NOT** an evaluation session!

NEXT STEPS…

CONTACT

lukasz.sienkiewicz1@pg.edu.pl

Online meeting 23.03.2026 18:00-20:00

MBA XVII ed. - Research Methods - Proposal Outline consultations 1 | Dołączanie do spotkania | Microsoft Teams

Online meeting 25.03.2026 18:00-20:00

MBA XVII ed. - Research Methods - Proposal Outline consultations 2 | Dołączanie do spotkania | Microsoft Teams

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HISTORY IS WISDOM
FUTURE IS CHALLENGE